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Software Engineering 2

Structure of a Design Document

Purpose of the DD

- Means of **communication**
 - Requirements analysts $\leftrightarrow$ Architects $\leftrightarrow$ Developers
- **Baseline** for implementation activities
- **Traceability**: mapping between requirements and components
- **Baseline** for integration and Quality Assurance
 - Identification of the order of implementation
 - Identification of the integration strategy (bottom-up vs top-down)
 - Supports verification and validation
- Refines the plan and previous **estimations**
 - size, cost, schedule

Reference structure for DD

(See the R&DD document)



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1 Introduction

Scope

Definitions, acronyms, abbreviations

Reference documents

Overview

2 Architectural Design

Overview: high-level components
and interactions

Component view

Deployment view

Component interfaces

Runtime view

Selected architectural styles
and patterns

Other design decisions

Reviews the domain and product, **summary of main architectural styles/choices** (e.g., N-tier / microservices, ...)

Describes contents and structure of the remainder of the DD

Informal view (free style notation), major interfaces

Components + interfaces:
component diagrams, class diagrams (detailed view)

Infrastructure: deployment diagram(s) including non-logical
elements (e.g., load balancer, firewall)

Details for each interface (name, signature, returned
objects, communication means, error handling)

Dynamics of the interactions: sequence diagrams (realization
of use cases)

Reference structure for DD

(See the R&DD document)



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3 User Interface Design

4 Requirements traceability

5 Implementation, Integration
and test Plan

6 Effort Spent

7 References

Overview of UIs, possibly mockups, may refine what's
in the RASD (if present)

Mapping between requirements and design elements

Order in which you plan to implement subsystems and
components as well as plan of the integration and test
of the integration



Homework

- Review the DD available on Webeep, direct link
 - https://webeep.polimi.it/pluginfile.php/1452976/mod_folder/content/0/ProjectToBeReviewed/DD.pdf
 - It refers to the assignment described in this document:
https://webeep.polimi.it/pluginfile.php/1452976/mod_folder/content/0/ProjectToBeReviewed/Assignment_RDD_2024-2025.pdf
- Answer to the questionnaire here (one submission per group):
 - <https://forms.office.com/e/Q8GEp6cpKL>
 - Groups: up to 3 students (same groups for the RASD questionnaire)
 - We will assign up to 1 point to clear and convincing answers
- **Deadline: Nov 19, 2025 at 23.59 (Rome time)**
- Answers will be used as basis for discussion during the lab of Nov 20, 2025



Homework — important note (repeat)

- Focus more on content rather than structure
- Content that is fully AI-generated will not be accepted and will receive a score of zero points.
 - Value your reasoning and expressive capabilities.
 - You are the ones who will build machines, not vice versa!
- The groups formed to submit this form should be the same as those for the RASD homework.